Antiderivatives and Riemann Sums

ANSWER KEY

Section 1: Basic Antiderivatives & Power Rule for Integration

1 Evaluate the indefinite integral $\int (6x^2 - 4x + 7)dx$.

a. $2x^3 - 2x^2 + 7x + C$ ✓

b. $3x^3 - 2x^2 + 7x + C$

c. $12x - 4 + C$

d. $2x^3 - 4x^2 + 7x + C$

Apply the power rule: $\int 6x^2 dx = 2x^3$, $\int -4x dx = -2x^2$, $\int 7dx = 7x$. Add constant of integration C.

2 Find $\int (\frac{3}{x} + \frac{1}{x^2})dx$ for $x > 0$.

a. $3\ln x - \frac{1}{x} + C$ ✓

b. $3\ln x + \frac{1}{x} + C$

c. $-\frac{3}{x^2} - \frac{2}{x^3} + C$

d. $3\ln x - \frac{1}{2x} + C$

$\int \frac{3}{x} dx = 3\ln|x|$ and $\int x^{-2} dx = -x^{-1} = -\frac{1}{x}$.

3 Find the general antiderivative of $f(x) = \sec^2 x + e^{3x}$.

a. $\tan x + \frac{1}{3}e^{3x} + C$ ✓

b. $\sec x \tan x + 3e^{3x} + C$

c. $\tan x + 3e^{3x} + C$

d. $2\sec x + \frac{1}{3}e^{3x} + C$

$\int \sec^2 x dx = \tan x$ and $\int e^{3x} dx = \frac{1}{3}e^{3x} + C$.

Section 2: Left and Right Riemann Sums

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4 Approximate $\int_0^4 x^2 dx$ using a Left Riemann sum with $n = 4$ equal subintervals.

a. 14 ✓

b. 30

c. 21.33

d. 16

$\Delta x = \frac{4-0}{4} = 1$. Subintervals: $[0, 1], [1, 2], [2, 3], [3, 4]$. Left endpoints: $0, 1, 2, 3$. Sum $= 1(0^2 + 1^2 + 2^2 + 3^2) = 1(0 + 1 + 4 + 9) = 14$.

5 What is the Right Riemann sum approximation for the above integral with $n = 4$?

a. 30 ✓

b. 14

c. 22

d. 21.33

Right endpoints: $1, 2, 3, 4$. Sum $= 1(1^2 + 2^2 + 3^2 + 4^2) = 1(1 + 4 + 9 + 16) = 30$.

6 If $f(x)$ is strictly increasing on $[a, b]$, what is the relationship between Left and Right Riemann sums and the true integral value I?

a. **Left Sum** $< I <$ **Right Sum** ✓

b. Right Sum $< I <$ Left Sum

c. Left Sum $=$ Right Sum

d. Cannot be determined without concavity

For an increasing function, rectangles using left endpoints underestimate the area while right endpoints overestimate the area: $\text{Left} < I < \text{Right}$.

Section 3: Midpoint & Trapezoidal Approximations

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7 Using the Trapezoidal Rule with $n = 4$ equal subintervals, approximate $\int_0^4 x^2 dx$.

a. 22 ✓

b. 21.33

c. 24

d. 20

Trapezoidal sum $= \frac{\text{Left} + \text{Right}}{2} = \frac{14 + 30}{2} = 22$.

8 If $f''(x) > 0$ (concave up) on $[a, b]$, how does the Trapezoidal Rule approximation compare to the exact integral?

a. **Trapezoidal Rule overestimates the true value** ✓

b. Trapezoidal Rule underestimates the true value

c. Trapezoidal Rule is exact

d. Depends on whether f is increasing

Chords connecting points on a concave up curve lie entirely above the curve, so trapezoids overestimate the true area.

9 Approximate $\int_0^2 x^3 dx$ using a Midpoint Riemann sum with $n = 2$ equal subintervals.

a. 3.5 ✓

b. 4.0

c. 4.5

d. 2.0

$\Delta x = 1$. Subintervals $[0, 1]$ and $[1, 2]$. Midpoints: $x_1^* = 0.5, x_2^* = 1.5$. Sum $= 1((0.5)^3 + (1.5)^3) = 0.125 + 3.375 = 3.5$.

Section 4: Limit of a Riemann Sum & Definite Integrals

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10 Which definite integral is equal to $\lim_{n\to\infty}\sum_{i=1}^{n}(1+\frac{2i}{n})^3\frac{2}{n}$?

a. $\int_1^3 x^3dx$ ✓

b. $\int_0^2 x^3dx$

c. $\int_1^2 (1+x)^3dx$

d. $\int_0^1 (1+2x)^3dx$

Here $\Delta x=\frac{2}{n}$, interval length $=2$. $x_i=1+i\Delta x\Rightarrow a=1, b=3$. The function is $f(x)=x^3$, giving $\int_1^3 x^3dx$.

11 Evaluate $\int_1^3 x^3dx$.

a. 20 ✓

b. 18

c. 26

d. 80

$[\frac{x^4}{4}]_1^3=\frac{81}{4}-\frac{1}{4}=\frac{80}{4}=20$.

12 Express $\int_0^5\sqrt{x}dx$ as a limit of a Right Riemann sum.

a. $\lim_{n\to\infty}\sum_{i=1}^{n}\sqrt{\frac{5i}{n}}\frac{5}{n}$ ✓

b. $\lim_{n\to\infty}\sum_{i=1}^{n}\sqrt{\frac{i}{n}}\frac{5}{n}$

c. $\lim_{n\to\infty}\sum_{i=1}^{n}\sqrt{5i}\frac{1}{n}$

d. $\lim_{n\to\infty}\sum_{i=1}^{n}\frac{5}{n}\sqrt{x_i}$

$\Delta x=\frac{5}{n}$ and right endpoints are $x_i=\frac{5i}{n}$, so the sum is $\lim_{n\to\infty}\sum_{i=1}^{n}\sqrt{\frac{5i}{n}}\frac{5}{n}$.

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